Mr. Scrooge

Automated Forex Trading Bot

Complete Trading Guide

Broker: OANDA | Strategy: Mean-Reversion Box Theory

Timeframe: 5-Minute Candles | Universe: ~24 Forex Pairs

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1. Overview

Mr. Scrooge is an automated mean-reversion forex trading bot that trades on OANDA using a box theory strategy. It watches for price to make an explosive move into a prior day's high/low zone, then catches the reversal snapback.

The bot runs on a 5-minute candle cycle with hourly and daily analysis layers. It uses margin-based position sizing (80% utilization), enforces a single concurrent position, and operates as a single instance via file locks.

2. Core Strategy: Box Theory + Mean Reversion

Each trading setup follows this sequence:

- **Daily Box:** Prior day's high (PDH) and low (PDL) form the box boundaries, filtered by percentile to trim single-bar spikes.
- **Box Breach:** Price breaches the box wall, entering a 'breach pending' state.
- **Sucker Move:** 3-5 consecutive candles drive into the zone (retail traders piling in).
- **Pitchfork Confirmation:** A reversal candle closes in the opposite direction, confirming the snapback.
- **Limit Order Entry:** Order placed at the pitchfork trigger price -- no market chasing.
- **Automated Exit:** Ride to profit with the 4-stage ratcheting stop system.

Named Signals

- **John Wick:** Hammer or inverted hammer candle at a zone extreme.
- **Power of Towers:** Full engulfing candle at a zone.

3. The 6-Gate Signal Filter

Every potential trade must pass through 6 gates in sequence:

Gate	What It Checks
Zone	Is price at a PDH/PDL magic line? (Tier 1/2/3)
Trend	Counter-trend entries blocked at weak zones (50-hour EMA)
ATR	Has price consumed >= 60% of daily ATR? (momentum check)
Sucker	Are there 3-5 directional candles into the zone?
Pitchfork	Has a reversal candle formed?
OTR	Is spread cost acceptable relative to box range?

Only when all gates pass does Scrooge enter a trade.

4. Position & Risk Management

Setting	Value

Max concurrent positions	1 (concentrated capital)
Stop loss	10 pips from fill
Take profit 1 (TP1)	+10 pips (ratchet trigger)
Take profit 2 (TP2)	+30 pips (full exit)
Min risk/reward	1.60
Daily loss halt	7.5% of equity
Max hold time	6 hours
Margin utilization	80% of cash balance

Ratchet Stop System (4 Stages)

- **Stage 1 (Entry):** SL sits 10 pips from fill price.
- **Stage 2 (At +5 pips):** SL moves to break-even.
- **Stage 3 (At +10 pips / TP1):** SL locks at entry + 10 pips.
- **Stage 4 (Above TP1):** Hybrid trailing stop (80% profit lock, 1.5x ATR floor).
- **Exit (At +30 pips / TP2):** OANDA's native TP closes the full position.

Harvester Daemon

A separate process runs every 30 seconds, monitoring positions for staggered partial closes at equity tiers (3%, 5%, 7%). It also closes stagnant profitable positions to free up capital. Per-pair pip overrides are configurable via harvester_config.json.

5. Spread & Cost Controls (OTR)

OTR = Overhead-to-Range (spread cost as a percentage of the box range):

OTR Level	Threshold	Action
Excellent	< 5%	Trade freely
Acceptable	5 - 15%	Normal rules apply
Marginal	15 - 25%	All 3 gates (zone + ATR + pitchfork) must align
Poor	> 25%	Skip entirely

Spread spikes greater than 2.5x baseline are also blocked automatically.

6. Session-Aware Trading

Scrooge knows which forex session is active and only trades relevant pairs. There is a 15-minute open wait after each session starts to let spreads settle.

Session	Hours (UTC)	Focus

Sydney	21:00 - 06:00	AUD/NZD, commodity crosses
Tokyo	00:00 - 09:00	JPY pairs, AUD/USD, NZD/USD
London	07:00 - 16:00	EUR/GBP majors (most active)
New York	13:00 - 22:00	USD pairs, London overlap

7. Instrument Universe

Base Pairs (Proven Net-Positive)

AUD_CAD, AUD_CHF, USD_JPY, EUR_CAD, CHF_JPY

Probation Pairs (Traded at 65% Size)

AUD_USD, EUR_GBP, EUR_JPY, GBP_AUD, GBP_CAD, GBP_JPY, NZD_USD

Total universe: ~24 forex pairs across majors, minors, and crosses. Universe screener operates in three modes: balanced (default), standard, and aggressive.

8. Opportunity Scoring (0-100)

Each setup receives a composite score. The highest-scored pair gets the 'ON DECK' badge in the dashboard.

Factor	Condition	Points
Zone Tier	Tier 1 / 2 / 3	30 / 20 / 10
ATR Strength	Strong / Valid / Weak	20 / 15 / 5
OTR Level	Excellent / Acceptable	15 / 10
Risk:Reward	>= 3.0 / >= 2.0 / >= 1.5	15 / 10 / 5
Signal Type	Named / Standard	10 / 5
Confidence	HIGH / MODERATE	5 / 3

9. Dashboard (Web UI)

The dashboard is accessible at localhost:8081 with HTTP Basic Auth. It auto-refreshes every 2 seconds and is built with pure HTML/CSS/JS (no frameworks).

Dashboard Tabs

- **Positions:** Live P&L;, ratchet stage, time in trade, entry price, current stop, TP levels.
- **Scanner:** All active pairs with box dimensions, ATR, OTR, spreads, session status, viability.
- **Signals:** Gate-by-gate pass/fail results for every pair. Entry/stop/TP prices, R:R ratio, rejection reasons.
- **Box States:** Per-pair state machine (ranging, breach pending, broken, new box, shelved). ON DECK badge for top candidate.

- **Config:** Edit any setting live in the browser. Applied next cycle, no restart needed.
- **Harvester:** Partial close tier thresholds, per-pair pip overrides, stagnant timeout settings.
- **Logs:** Filterable real-time log viewer with separate streams for decisions, trades, and runtime.

Status Bar

Displays equity, daily P&L;, daily loss %, positions open, active sessions, and weekend indicator.

10. All Configuration Settings

All settings live in main.py's CONFIG dict and can be overridden live via the dashboard (saved to config.json). Organized by category:

Position Sizing

Setting	Default	Description
margin_utilization_pct	0.80	80% of cash balance as margin
max_concurrent_positions	1	Single position at a time

Stop Loss / Take Profit

Setting	Default	Description
sl_pips	10	Stop loss distance from fill price
tp1_pips	10	Ratchet trigger level
tp2_pips	30	Full exit level

Timing

Setting	Default	Description
scan_interval_seconds	300	5-minute scan loop
rescan_interval_minutes	30	Universe rescan frequency
open_wait_minutes	15	Block entries post-session-open
max_hold_minutes	360	6-hour max position hold
hourly_green_exit_enabled	True	Close profitable positions hourly

Risk Gates

Setting	Default	Description
min_risk_reward	1.60	Minimum R:R to enter
max_daily_loss_pct	0.075	7.5% daily equity halt
min_net_profit_pips	7	Floor profit in pips

min_efficiency	2.0	Net profit / round-trip cost

OTR Thresholds

Setting	Default	Description
otr_excellent	0.05	< 5% spread cost
otr_acceptable	0.15	5-15% spread cost
otr_marginal	0.25	15-25% spread cost

Box & Zone

Setting	Default	Description
zone_proximity_pct	0.0025	0.25% tolerance to be 'in zone'
atr_period	14	ATR lookback period
atr_valid_pct	0.60	60% ATR = valid setup
atr_minimum_pct	0.15	Below 15% = dead market
spread_spike_threshold	4.0	Spread must stay < 4x baseline
sl_floor_pct	0.0050	0.50% minimum SL distance

Universe

Setting	Default	Description
instrument_tiers	majors	Which tier of pairs to scan
universe_mode	balanced	Session-aware pair selection mode
apply_probation_size	True	Reduce size on probation pairs
probation_size_multiplier	0.65	65% of normal for probation

11. Architecture

Mr. Scrooge is structured as follows:

```
mr-scrooge/
main.py Boot, 5-min loop, daily reset, ranking
core/
oanda_client.py OANDA v20 REST wrapper
execution_router.py Order routing (limit + market fallback)
fee_calculator.py Spread cost & OTR calculations
strategy/
zone_detector.py PDH/PDL box, percentile filter, gaps
signal_engine.py 6-gate evaluation, pitchfork, scoring
box_state_machine.py Per-pair box state tracking
```

```
universe_screener.py Session-aware pair selection
portfolio/
trade_manager.py Position lifecycle, ratchet, P&L
harvester/
main.py Profit daemon (partials, trailing)
dashboard.py Web server (stdlib HTTP)
dashboard.html UI (pure HTML/CSS/JS)
```

Dependencies are minimal: requests, pandas, and python-dotenv. The dashboard uses no external JS libraries. Deployed on EC2 via systemd with auto-restart on crash (15-second delay).

12. Key Algorithms

Sucker Move Detection

- 3-5 consecutive bars moving in one direction
- Must consume >= 60% of daily ATR (valid momentum)
- Must terminate at a known magic line (zone)
- Detects the explosive shakeout that precedes reversal

Pitchfork Confirmation

- Reversal candle: first bar breaking direction (closes opposite)
- Entry on break of reversal high (long) or low (short)
- Timeframe: 5M normal, 1H if volatility > 2x hourly ATR
- Named variations: John Wick (hammer) and Power of Towers (engulfing)

Trend Regime Filter

- 50-hour EMA on 1H bars determines uptrend/downtrend
- Counter-trend entries blocked at Tier 3/4 zones
- Tier 1/2 zones override trend (strong enough to fight)

Box State Machine

Each pair tracks through states: IN_BOX_RANGING -> BOX_BREACH_PENDING -> BOX_BROKEN_RECALCULATING -> NEW_BOX_STANDBY -> SHELVED

- Box age confidence decays through session
- Alternative box drawing when original breaks
- Return-to-box setup detection

13. Lessons Learned

- 30-minute rolling box too unstable -- reverted to daily PDH/PDL.
- Fixed-pip stops (from fill price) more consistent than zone-anchored stops.
- Sizing must use base_price_usd x margin_rate, not current_price x margin_rate.

- Ratchet TP1 at 10 pips, not 5 (too tight at 5 pips).
- Must recalculate SL/TP from actual fill price, not trigger price.
- Full 80% margin at all times is simpler than dynamic decompressors.
- Use balance (cash) for sizing, not NAV (includes unrealized P&L;).

14. TL;DR Summary

Scrooge watches for price to slam into yesterday's high or low, waits for a reversal candle to confirm the snapback, enters via limit order, then manages the trade with an automated 4-stage ratcheting stop system -- targeting 30 pips profit with a 10-pip stop. One position at a time, session-aware, with a 7.5% daily loss circuit breaker.